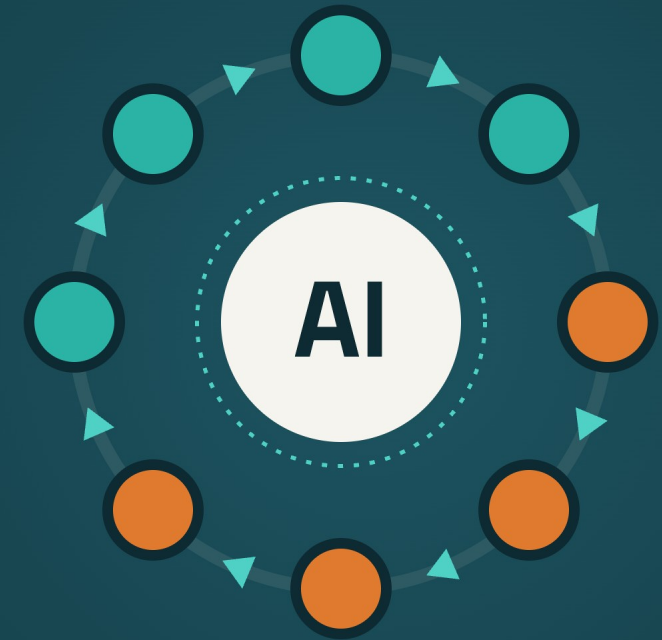


Integrated Data Center & Urban Cooling System Using Greywater

Greywater → cools AI → cools the city



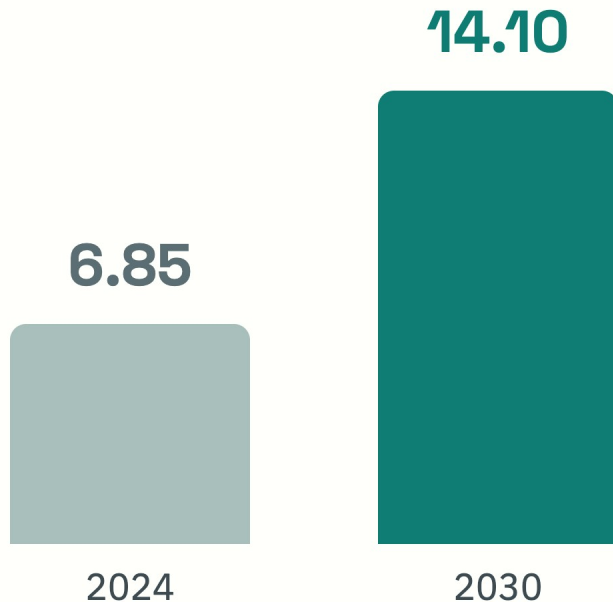
Done By: Walae Darif S24109318, Asmaa Jama S24109128, Nafisa Mahmudalhasan S24109378, Basma Elfaki S24109310

Course: introduction to ai

Three growing problems

Wastewater will double

SAUDI ARABIA · MILLION m³ / DAY



AI runs hot

More compute → more heat → more water



Streets trap heat

Pavements store the desert sun



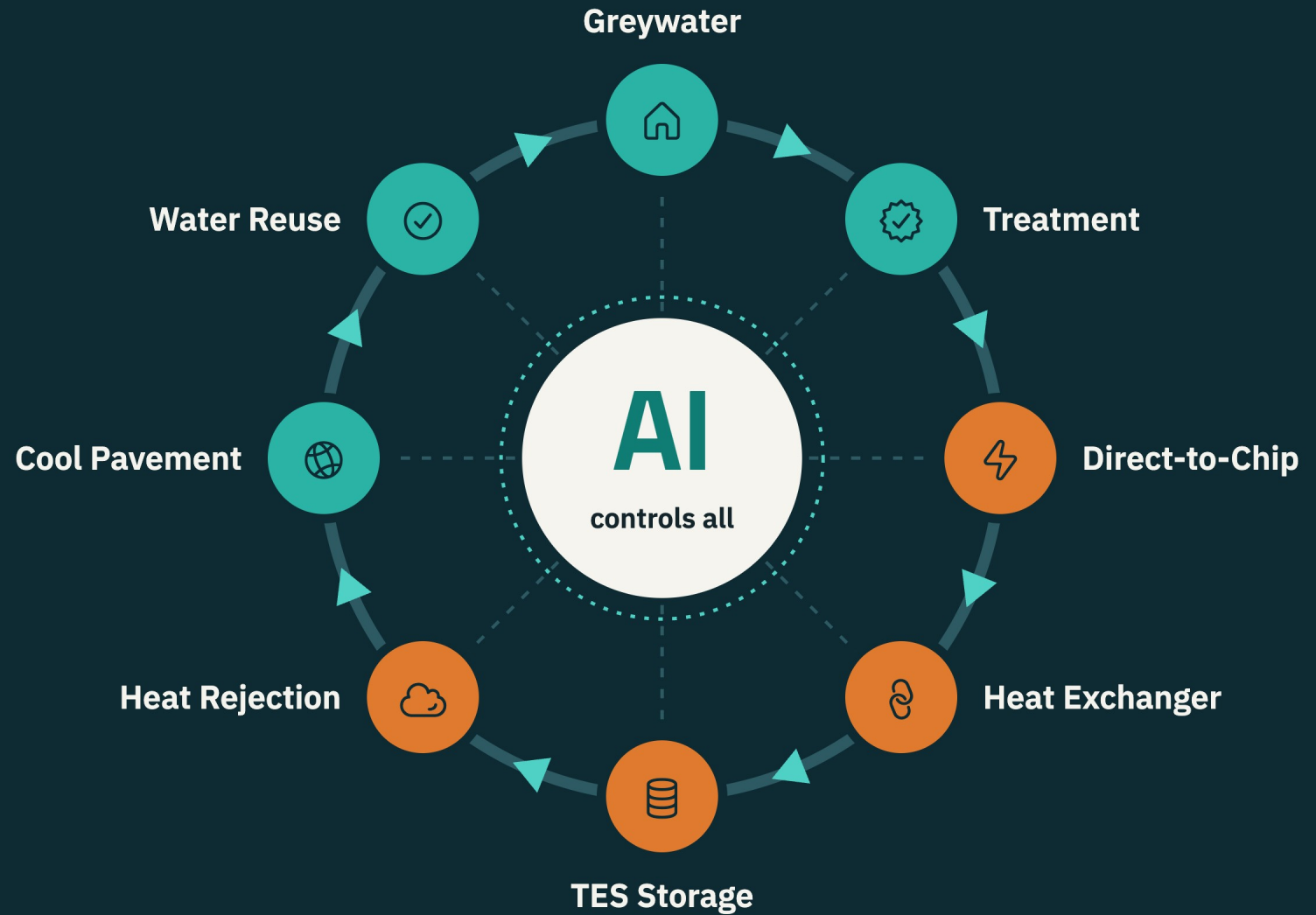
Solved separately

We connect all three in one loop

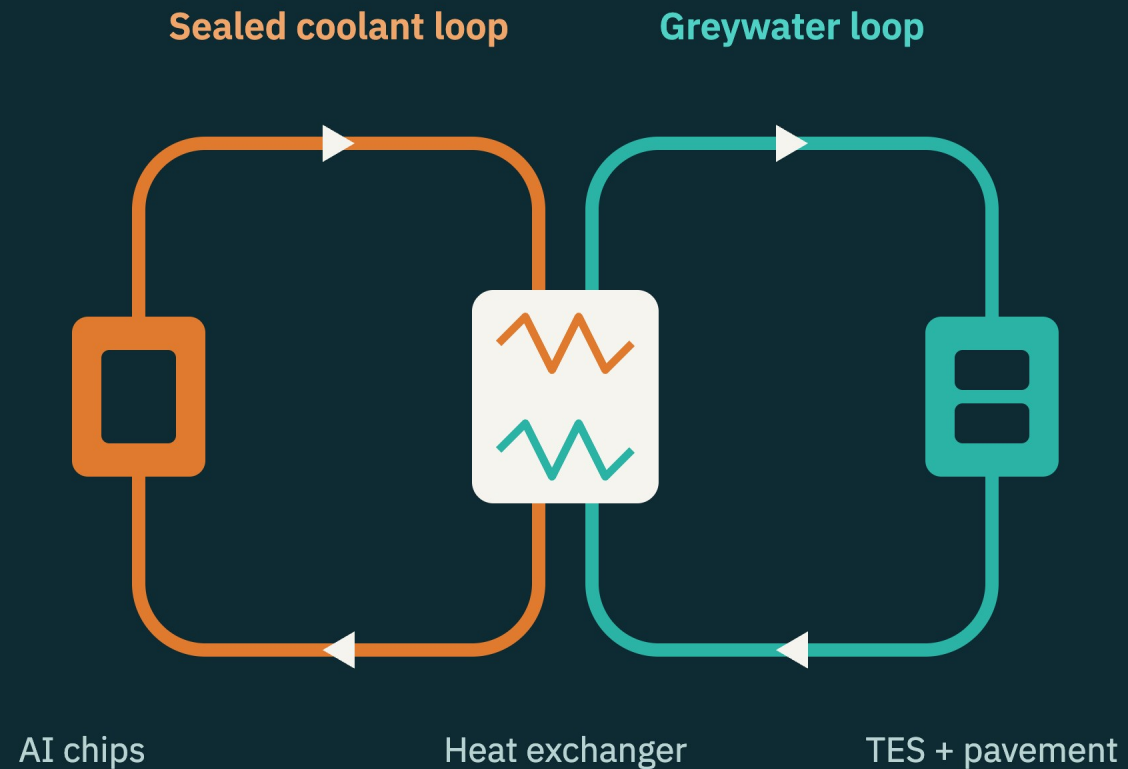
HOW IT WORKS

One circular loop

8 steps. 1 AI brain. Heat moves, water stays.



Direct-to-chip: the new standard



Greywater never touches the chips. Only heat crosses over.

125M+ L

water saved per data center per year · Microsoft, 2026 sites

300×

better water efficiency · NVIDIA liquid-cooled AI racks

AI: the brain of the loop

IT READS

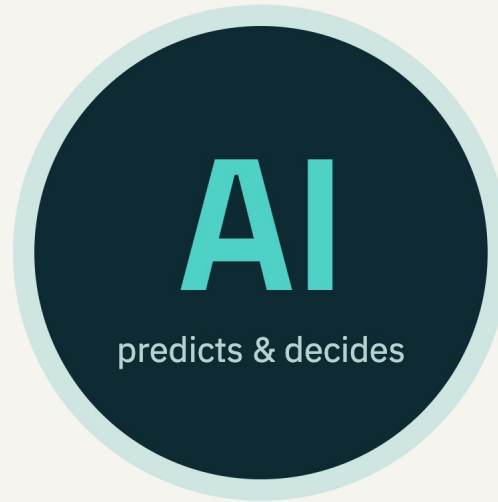
Computing load

Weather

Pavement temperature

Water quality

Storage capacity



IT CONTROLS

Pump speed & flow

Heat to TES or pavement

UV & dosing

Leak & fault alerts

Predict → Decide → Control → Learn

Why Saudi Arabia, why now

70%

water reuse target by 2030

2026

Saudi Year of AI

**100
MW**

AI data center rising at
NEOM Oxagon

Vision 2030: water · AI · smart cities, in one system

THE VISION

Used water shouldn't be waste. It should cool our AI and our cities.

AI data centers

Smart buildings

Smart cities

Water utilities

Thank you. Questions?